

Algorithms Problem Solving

Level 3

Transpose Matrix



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Definition

 Transpose Matrix

 Program creates a 3×3 ordered matrix and performs a transpose operation (swapping rows with columns).

Explanation

 Matrix Transposition

 Swaps indices so element at row `i` and column `j` moves to row `j` and column `i`.

Procedures

1. Create a 3×3 matrix.
2. Fill matrix with ordered numbers from 1 to 9.
3. Print original matrix via PrintMatrix().
4. Perform transposition in TransposeMatrix().
5. Display transposed matrix via PrintTransposedMatrix().

💡 Example

The following is a 3x3 ordered matrix:

1	2	3
4	5	6
7	8	9

The following is the transposed matrix:

1	4	7
2	5	8
3	6	9

➡ Result

Rows become columns, and columns become rows.

Note

◆ In the transpose operation, the element at `[i][j]` becomes `[j][i]` in the transposed matrix.

◆ Example:

Original: `matrix[1][0] = 4`

Transposed: `transposed[0][1] = 4`

◆ `TransposeMatrix()` creates a new matrix named `transposed`, so the original `matrix` is not modified.

◆ `PrintTransposedMatrix()` can print the transposed values directly without changing the original matrix.

Comparison

Original Matrix

- ✓ Rows remain rows.
- ✓ Columns remain columns.
- ✓ Example:

1	2	3
4	5	6
7	8	9

Transposed Matrix

- ✓ Rows become columns.
- ✓ Columns become rows.
- ✓ Example:

1	4	7
2	5	8
3	6	9

✦ Summary

- ✓ Performs classic matrix transposition by interchanging row indices with column indices (`matrix[i][j]` to `transposed[j][i]`).
- ✓ Creates a separate transformed matrix structure while keeping the source matrix intact.
- ✓ Demonstrates fundamental matrix manipulation essential for advanced linear algebra and graphics algorithms.

Thank you 😊